

EXP-4

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Subject Code: 24CSP-310

Subject Name : Advanced Database Management System

A)

(i) You are given a table `tbl_happiness` that contains the ranking of happiness for different countries. Your task is to retrieve all the records from the table **while ensuring that "India" and "Sri Lanka" appear at the top, followed by the rest of the countries in their original ranking order.**

CODE:

```
--A.  
CREATE TABLE tbl_happiness  
(  
    sno INT PRIMARY KEY,  
    rankings INT,  
    country VARCHAR(50)  
);  
  
INSERT INTO tbl_happiness  
VALUES  
(1,1,'Finland'),  
(2,2,'Denmark'),  
(3,3,'Iceland'),  
(4,4,'Israel'),  
(5,5,'Netherlands'),  
(6,6,'Sweden'),  
(7,7,'Norway'),  
(8,126,'India'),  
(9,128,'Sri Lanka');
```

```
SELECT *,
CASE
    WHEN country = 'India' THEN 1
    WHEN country = 'Sri Lanka' THEN 2
    ELSE 3
END AS DERIVED_RANK
FROM tbl_happiness
ORDER BY DERIVED_RANK, RANKINGS;
```

OUTPUT SCREENSHOTS:

	sno [PK] integer	rankings integer	country character varying (50)	derived_rank integer
1	8	126	India	1
2	9	128	Sri Lanka	2
3	1	1	Finland	3
4	2	2	Denmark	3
5	3	3	Iceland	3
6	4	4	Israel	3
7	5	5	Netherlands	3
8	6	6	Sweden	3
9	7	7	Norway	3

B)

(ii) Make all the rankings in descending order except **India** and **Sri Lanka**.

CODE:

```
CREATE TABLE tbl_happiness
(
    sno INT PRIMARY KEY,
    rankings INT,
    country VARCHAR(50)
);
```

```
INSERT INTO tbl_happiness
VALUES
(1,1,'Finland'),
(2,2,'Denmark'),
(3,3,'Iceland'),
(4,4,'Israel'),
(5,5,'Netherlands'),
(6,6,'Sweden'),
(7,7,'Norway'),
(8,126,'India'),
(9,128,'Sri Lanka');
```

```
SELECT *,
CASE
    WHEN country = 'India' THEN 1
    WHEN country = 'Sri Lanka' THEN 2
    ELSE 3
END AS DERIVED_RANK
FROM tbl_happiness
ORDER BY DERIVED_RANK, RANKINGS DESC;
```

OUTPUT SCREENSHOTS:

	sno [PK] integer	rankings integer	country character varying (50)	derived_rank integer
1	8	126	India	1
2	9	128	Sri Lanka	2
3	7	7	Norway	3
4	6	6	Sweden	3
5	5	5	Netherlands	3
6	4	4	Israel	3
7	3	3	Iceland	3
8	2	2	Denmark	3
9	1	1	Finland	3

C)

A company maintains employee information in a single table where each employee may report to another employee acting as their manager. Prepare a report that displays the details of every employee who supervises at least one other employee. The report should include the manager's employee ID, name, total number of direct reports, and the average age of those reporting employees. The average age should be rounded to the nearest whole number and the final result should be sorted by employee ID.

CODE SCREENSHOTS:

```
CREATE TABLE Employees
(
    employee_id INT PRIMARY KEY,
    name VARCHAR(50),
    reports_to INT,
    age INT
);

INSERT INTO Employees (employee_id, name, reports_to, age)
VALUES
(9, 'Hercy', NULL, 43),
(6, 'Alice', 9, 41),
(4, 'Bob', 9, 36),
(2, 'Winston', NULL, 37);

SELECT
    e1.employee_id,
    e1.name,
    COUNT(e2.employee_id) AS reports_count,
    ROUND(AVG(e2.age), 0) AS average_age
FROM Employees e1
JOIN Employees e2
    ON e1.employee_id = e2.reports_to
GROUP BY e1.employee_id, e1.name
ORDER BY e1.employee_id;
```

OUTPUT SCREENSHOTS:

	employee_id [PK] integer	name character varying (50)	reports_count bigint	average_age numeric
1	9	Hercy	2	39

LEARNING OUTCOMES:

- Learned to perform custom sorting using ORDER BY and CASE statements.
- Gained experience in using self-joins to analyze hierarchical employee data.
- Learned to calculate percentages using conditional aggregation.

- Developed skills in combining multiple SQL concepts to solve complex queries.
- Improved proficiency in writing optimized SQL queries for real-world scenarios.

CONCLUSION:

Successfully implemented advanced SQL queries involving custom sorting, self-joins, conditional aggregation, and percentage calculations to produce accurate results.